**EXP 25 – Monkey Banana Problem**

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**Aim**

To write a Prolog program that models the Monkey–Banana problem using state transitions and recursive reasoning, and to determine whether the monkey can reach and obtain the banana.

**Algorithm**

1. Represent the monkey's state as: **state(MonkeyPosition, MonkeyStatus, BoxPosition, HasBananaStatus)**
2. Define possible actions:
   - **walk(P1, P2):** Monkey walks from position P1 to P2.
   - **push(P1, P2):** Monkey pushes the box from P1 to P2.
   - **climb:** Monkey climbs onto the box.
   - **grasp:** Monkey grasps the banana when on the box under the banana.
3. Define move(CurrentState, Action, NextState) rules for each action.
4. Define canget(State) which checks if the monkey already has the banana.
5. If not, recursively try all possible moves until the banana is obtained.

**Source Code**

prolog

```
move(state(middle, onbox, middle, hasnot), grasp,
   state(middle, onbox, middle, has)).


move(state(P, onfloor, P, H), climb,
   state(P, onbox, P, H)).
```

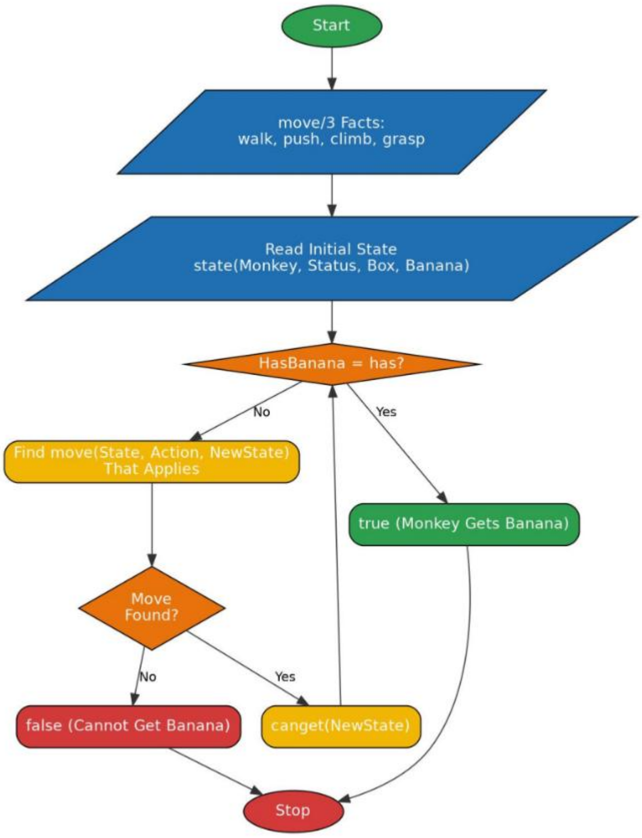
```
move(state(P1, onfloor, P1, H), push(P1, P2),
    state(P2, onfloor, P2, H)).

move(state(P1, onfloor, B, H), walk(P1, P2),
    state(P2, onfloor, B, H)).

canget(state(_, _, _, has)) :- !.

canget(State1) :-
    move(State1, _, State2),
    canget(State2).
```

**Flowchart:**




**Output:**

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [monkey].
true.

?- canget(state(atdoor, onfloor, atwindow, hasnot)).
true .

?- canget(state(middle, onbox, middle, hasnot)).
true.
```

**Result**

The program successfully simulates the monkey's actions and verifies that through walking, pushing the box, climbing, and grasping, the monkey can eventually obtain the banana. The recursive search confirms that the final state where the monkey "has" the banana is reachable.